S5 - Glossary

Terminology used in the occupancy secondary analysis

Article type: Supplementary Material

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Abstract Terminology for the Apollo propofol occupancy article. Entries distinguish (1) **Implemented** — the exact computation; (2) **Interpretation** — the most defensible empirical reading; (3) **Not established** — theoretical mappings not directly confirmed here.

A - D

Baseline-fixed chart - **Implemented**: eight `features_raw` columns robust-z scored with that participant's pre-drug median and MAD. **Interpretation**: a within-person reference that can see occupancy. **Not established**: identification of a latent neural manifold.

Caliper - Median baseline leave-one-out nearest-neighbour Euclidean distance in the baseline-fixed chart; greedy matches farther than the caliper are rejected.

Common support - Overlap between later-phase and baseline clouds that permits matching. Collapse of common support is why C3 is NOT_TESTABLE here.

d_z - Paired Cohen's d: mean of within-subject differences divided by their sample SD.

E - L

Eligibility - Finite occupancy features and observed $\Delta t \in [3.8, 4.2]$ s on a one-step source window.

First / later protocol sedation - `datainfo` orders 2 and 3 (mild / moderate in the source paper). Later protocol sedation is not LOR.

G0–G4 - Preregistration gates: inventory, labels, support, independent code review, lock. G0–G2 closed; G3/G4 not completed. All tests exploratory.

LOR / ROR - Loss / return of behavioural responsiveness from the Chennu binomial-CI rule on $k/40$, occupancy-blind. Not consciousness.

M - R

Matched state (NDT-C3 analogue) - Increment speed/spread compared after pairing later and baseline windows at similar z. **Implemented** here with greedy Euclidean matching. **Not established**: a change in the latent future law.

MNPS - Exported three-dimensional neural-manifold coordinate. Occupancy does not use it as the primary state.

NDT-C1 / C3 - Present-state geometry versus conditional future law at matched state (Langell, 2026).

Occupancy - Median $\|z\|_2$ of eligible unmatched windows in the baseline-fixed chart. **Interpretation**: relocation away from the awake neighbourhood. **Not established**: anaesthetic depth or unconsciousness.

Protocol phase - Cambridge `datainfo` column 2: baseline, first sedation, later sedation, recovery. Not inferred from file order.

S - Z

Sign test - Exact two-sided binomial test on the sign of non-zero paired differences.

Subject-anchored (export) - Recording-local MNPS scaling in the HDF5 bundle; ComBat disabled. Distinct from the article-side baseline-fixed occupancy frame.

Unresponsive (package alias) - Legacy name for protocol order 3. Not used as LOR.

Bibliography

Langell, R. (2026,). *Noetic Diffusion Theory: A Rhythmic, Geometric Framework for Neural State Reconstruction and Conscious Correlates*. Zenodo. https://doi.org/10.5281/zenodo.20112950